

EC824A

Asymptotic Theory of Testing

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Testing

- Suppose we begin with a problem of testing a simple null hypothesis

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$$

- For specific alternative, $\theta = \theta_1$, “know” the best test of H_0 has critical regions,

$$\left\{ z \mid \frac{L(\theta_1, z)}{L(\theta_0, z)} > c \right\}, \quad (\text{Neyman-Pearson Lemma})$$

- When alternative not simple, can get uniformly most powerful test using critical regions

$$\left\{ z \mid \frac{L(\hat{\theta}_n, z)}{L(\theta_0, z)} > c \right\}$$

where $\hat{\theta}_n$ denotes the MLE optimum

- Finite n , exact distribution of likelihood ratio difficult – study asymptotics

1 Testing Introduction

2 Multivariate Testing

3 Composite Hypothesis

Testing Introduction – Example

- Consider the linear model with normally distributed errors and known variance,

$$y = x'\beta + u, \quad u \sim N(0, \sigma^2 I_n)$$

- Structure of hypothesis test is to “guess”, or question, a supposed β_0 , $H_0 : \beta = \beta_0$
- Likelihood and likelihood ratio:

$$L(\beta, y) = (2\pi\sigma^2)^{-n/2} \exp \left\{ - \sum (y_i - x_i\beta)^2 / 2\sigma^2 \right\}$$

$$\lambda := L(\hat{\beta}, y) / L(\beta_0, y) = \exp \left\{ \frac{1}{2\sigma^2} (\hat{\beta} - \beta_0)' X' X (\hat{\beta} - \beta_0) \right\}$$

$$2 \log \lambda = (\hat{\beta} - \beta_0)' X' X (\hat{\beta} - \beta_0) / \sigma^2$$

- Thus, under H_0 , $2 \log \lambda \sim \chi_p^2$, and can construct a decision rule for exact level α test
- When σ^2 unknown, estimate, and use $F_{p, n-p}$ for critical values

Testing Introduction – χ^2

- Remember, for $Z_1, \dots, Z_p$ independently distributed as $N(0, 1)$, we have that $\sum_{k=1}^p Z_k^2 \sim \chi_p^2$, i.e. a Chi-square distribution with p degrees of freedom
- What if Z_i 's not centred around zero? E.g., if a test statistic of parameters in the linear model when the null H_0 is false, the test statistic is not centred at zero.

Definition 1

- $Z_1, \dots, Z_p$ independently distributed as $Z_k \sim N(\xi_k, \sigma^2)$
- $\sum_{k=1}^p Z_k^2 / \sigma^2 \sim \chi_p^2(\delta)$: non-central χ^2 with p d.o.f., non-centrality measure $\delta = \sigma^{-2} \sum_k \xi_k^2$
- Generally, if p -vector $Z \sim N(\xi, \Omega)$, then $Z' \Omega^{-1} Z \sim \chi_p^2(\delta)$, $\delta = \xi' \Omega^{-1} \xi$

Testing Introduction – Example

- Natural question: does our test reject H_0 when it is not true?
- “Power”: probability to reject null given H_1 true, i.e. $1 - b$ where b is Type II error
- If H_0 false: $\beta = \beta_1 \neq \beta_0$, $\hat{\beta} - \beta_0 \sim N(\beta_1 - \beta_0, \sigma^2(X'X)^{-1})$, and $2 \log \lambda \sim \chi_p^2(\delta_n)$, with,

$$\delta_n = (\beta_1 - \beta_0)' X' X (\beta_1 - \beta_0) / \sigma^2$$

- What happens δ_n as $n \rightarrow \infty$? $D := \lim_{n \rightarrow \infty} n^{-1} X' X \gg 0$ (p.d.), and for large n :

$$\delta_n \approx n(\beta_1 - \beta_0)' D (\beta_1 - \beta_0) / \sigma^2$$

hence, $\mathbb{E} \chi_p^2(\delta_n) = p + \delta_n \rightarrow \infty$, $\mathbb{V} \chi_p^2(\delta_n) = 2p + 4\delta_n \rightarrow \infty$, i.e. mean and variance diverge

- Hence, under H_1 , test statistic diverges and Type II error converges to zero, i.e. power converges to 1, “test has power”, test is consistent, etc.

Testing Introduction – Example

- BUT: this was under a known, fixed, alternative hypothesis
- To realistically study power, consider “sequence of local alternatives”

$$H_0 : \beta = \beta_0 \quad \text{vs.} \quad H_1 : \beta = \beta_1 = \beta_0 + \xi/a_n,$$

$a_n \rightarrow \infty$ at rate to stabilise power, e.g. $a_n = \sqrt{n}$:

$$2 \log \lambda = (\hat{\beta} - \beta_0)' X' X (\hat{\beta} - \beta_0) / \sigma^2 \xrightarrow{d} \chi_p^2(\delta_0)$$

where $\delta_n = n^{-1} \xi' X' X \xi \rightarrow \xi' D \xi =: \delta_0$

- Then, if there is asymptotically non-negligible difference, i.e. $\xi \neq 0$, power can be discerned, without it trivially converging to 1 or 0.

Testing Introduction – General

- Let's look at a more general likelihood setting
- Let $\ell(\theta)$ be loglikelihood. Expand around $\hat{\theta}$

$$\ell(\theta_0) = \ell(\hat{\theta}) + (\theta_0 - \hat{\theta})\ell'(\hat{\theta}) + \frac{1}{2}(\theta_0 - \hat{\theta})^2\ell''(\theta^*)$$

where θ^* between θ_0 , and $\hat{\theta}$. Let λ be the likelihood ratio,

$$\begin{aligned} 2\log \lambda &= 2(\ell(\hat{\theta}) - \ell(\theta_0)) \\ &= 2\{-(\theta_0 - \hat{\theta})\ell'(\hat{\theta}) - \frac{1}{2}(\theta_0 - \hat{\theta})^2\ell''(\theta^*)\} \\ &= -(\hat{\theta} - \theta_0)^2\ell''(\theta^*). \end{aligned}$$

- Since $\hat{\theta} \xrightarrow{P} \theta_0$, $n^{-1}\ell''(\theta^*) \xrightarrow{P} -\mathcal{I}(\theta_0)$ and $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \mathcal{I}(\theta_0)^{-1})$ we have

$$2\log \lambda = -n(\hat{\theta} - \theta_0)^2 \left(\frac{1}{n}\ell''(\theta^*) \right) \xrightarrow{d} \chi_1^2(0)$$

Testing Introduction – Wald

- Wald test:

$$W(\theta) = \frac{(\hat{\theta} - \theta)^2}{\text{Var}(\hat{\theta})}$$

- Wald test has same distribution as $2 \log \lambda$ under H_0 or H_1

$$W(\theta_0) = \frac{(\hat{\theta} - \theta_0)^2}{\text{Var}(\hat{\theta})} \xrightarrow{d} \chi_1^2(0), \quad W(\theta_0) = \frac{(\hat{\theta} - \theta_0)^2}{\text{Var}(\hat{\theta})} \xrightarrow{d} \chi_1^2(\delta)$$

where $\delta = \xi' \text{AsyVar}(\sqrt{n}\hat{\theta})^{-1} \xi$

- (Asymptotic variance from linear model above was $n^{-1} \sigma^2 [\mathbb{E} x_i x_i']^{-1}$)

Testing Introduction – Lagrange Multiplier

- The Score, or Lagrange Multiplier (LM), test based on statistic under null:

$$S(\theta_0) = \frac{U(\theta_0)^2}{\mathcal{I}(\theta_0)} \quad \text{where,} \quad U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta} =: \ell'(\theta)$$

- Checks gradient ($U(\theta_0)$) relative to variance ($\mathcal{I}(\theta_0)$)
- When calculated, larger $S(\hat{\theta})$ implies $U(\hat{\theta})$ is further from 0, hence sub-optimum
- Expand $U(\theta_0)$ around MLE solution $\hat{\theta}$

$$\ell'(\theta_0) = \ell'(\hat{\theta}) + (\theta_0 - \hat{\theta})\ell''(\hat{\theta}) + \frac{1}{2}(\theta_0 - \hat{\theta})^2\ell'''(\theta^*)$$

- Since $\hat{\theta}$ is MLE, $\ell'(\hat{\theta}) = 0$. Take $\xi := \sqrt{n}(\hat{\theta} - \theta_0)$, i.e. local alternative, then,

$$\begin{aligned} \sqrt{n}\frac{1}{n}\ell'(\theta_0) &= \sqrt{n}(\hat{\theta} - \theta_0)(-\ell''(\hat{\theta})/n) + \frac{1}{2}\sqrt{n}(\theta_0 - \hat{\theta})^2\ell'''(\theta^*)/n \\ &\xrightarrow{P} \sqrt{n}(\hat{\theta} - \theta_0)\mathcal{I}(\theta_0) \xrightarrow{d} N(\mathcal{I}(\theta_0)\xi, \mathcal{I}(\theta_0)) \end{aligned}$$

Testing Introduction – Lagrange Multiplier

- Have, $(\sqrt{n}/n)\ell'(\theta_0) \xrightarrow{d} N(\mathcal{I}(\theta_0)\xi, \mathcal{I}(\theta_0))$. Hence, for $\delta = \xi'\mathcal{I}(\theta_0)\xi$

$$\frac{1}{n}(\ell'(\theta_0))'\mathcal{I}(\theta_0)^{-1}\ell'(\theta_0) \xrightarrow{d} \chi_1^2(\delta) \quad \text{equivalently} \quad (\ell'(\theta_0))'(-\frac{1}{n}\ell''(\theta_0))^{-1}\ell'(\theta_0) \xrightarrow{d} \chi_1^2(\delta)$$

- Hence, asymptotically equivalent to Wald and “general” under H_0 or H_1
- Why Lagrange? Consider,

$$\max \ell(\theta) - \lambda(\theta - \theta_0)$$

with FOC: $\ell'(\theta_0) = \lambda$. Hence, test based on score function is the same as test based on lagrange multiplier

1 Testing Introduction

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3 Composite Hypothesis

Multivariate Testing

- $\Theta \in \mathbb{R}^p$, and $H_0 : \theta = \theta_0$ and $H_n : \theta = \theta_n = \theta_0 + \xi/\sqrt{n}$.

Theorem 2 (Asymptotic Equivalence of the Trinity)

Under the Lehmann Conditions, and H_n , the test statistics

$$W = n(\hat{\theta}_n - \theta_0)' \mathcal{I}(\theta_0)(\hat{\theta}_n - \theta_0)$$

$$LR = 2(\ell(\hat{\theta}_n) - \ell(\theta_0))$$

$$LM = n^{-1}(\nabla \ell(\theta_0))' \mathcal{I}(\theta_0)^{-1} \nabla \ell(\theta_0)$$

all converge in distribution to $\chi_p^2(\delta)$, with $\delta = \xi' \mathcal{I}(\theta_0) \xi$

- Lehmann conditions are technical regularities we can safely take for granted

Multivariate Testing

Proof.

Under H_n , $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\xi, \mathcal{I}(\theta_0)^{-1})$ so $W \xrightarrow{d} \chi_p^2(\delta)$. Expanding $\ell(\cdot)$ around $\hat{\theta}_n$ as in the scalar case yields $LR \xrightarrow{d} \chi_p^2(\delta)$. Expanding $\nabla \ell(\cdot)$ around $\hat{\theta}_n$ as in the scalar case yields

$$n^{-1/2} \nabla \ell(\theta_0) = n^{-1/2} \nabla \ell(\hat{\theta}_n) + (-\nabla^2 \ell(\hat{\theta}_n)/n) \sqrt{n}(\hat{\theta}_n - \theta_0)' + o_p(1) \xrightarrow{d} N(\mathcal{I}(\theta_0)\xi, \mathcal{I}(\theta_0))$$

so $LM \xrightarrow{d} \chi_p^2(\delta)$. ■

- Exercise: show your working for Trinity (Hint: for LM, define $\tilde{\xi} = \mathcal{I}(\theta_0)\xi$, and proceed by using χ^2 fact: for $Z \sim N(\tilde{\xi}, \Omega)$, $Z'\Omega^{-1}Z \sim \chi^2(\tilde{\delta})$ where $\tilde{\delta} = \tilde{\xi}'\Omega^{-1}\tilde{\xi}$)

1 Testing Introduction

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Composite Hypothesis

- With many parameters, usually only interested in inference in small subset of them
- Partition $\theta = (\theta_1, \theta_2)$. Consider $H_0 : \theta_1 = \theta_{10}$ vs. $H_n : \theta_1 = \theta_{10} + \xi/\sqrt{n}$, θ_2 unrestricted
- Let $\hat{\theta}_n$ and $\tilde{\theta}_n$ denote the unrestricted and restricted mle's respectively
- $\tilde{\theta}_n = (\theta_{10}, \hat{\theta}_2)$, fixes θ_{10} , leaves nuisance parameters θ_2 unrestricted
- Partition $\nabla \ell = (s_1(\theta), s_2(\theta))'$ and $\mathcal{I}(\theta) = (\mathcal{I}_{ij}(\theta))$. Let $\mathcal{I}^{ij}$ denote the ij block of $\mathcal{I}^{-1}$.

Theorem 3

Under Conditions for Lehmann–Scheffé theorem, and H_n ,

$$W = n(\hat{\theta}_1 - \theta_{10})' \mathcal{I}^{11}(\theta_0)^{-1} (\hat{\theta}_1 - \theta_{10})$$

$$LR = 2(\ell(\hat{\theta}_n) - \ell(\tilde{\theta}_n))$$

$$LM = n^{-1} s_1(\tilde{\theta}_n)' \mathcal{I}^{11}(\tilde{\theta}) s_1(\tilde{\theta}_n)$$

all converge in distribution to $\chi_q^2(\xi' \mathcal{I}^{11}(\theta_0)^{-1} \xi)$ where $q = \dim \theta_1 \leq p$.

Composite Hypothesis

Proof.

- Wald: know $\sqrt{n}(\hat{\theta}_1 - \theta_{01}) \xrightarrow{d} N(\xi, \mathcal{I}^{11}(\theta_0))$, hence $W \xrightarrow{d} \chi_q^2(\delta)$ with $\delta = \xi' \mathcal{I}^{11}(\theta_0)^{-1} \xi$
 - LR and LM: defer - the next theorem is more general
-
- Note: $\mathcal{I}^{11} = (\mathcal{I}_{11} - \mathcal{I}_{12} \mathcal{I}_{22}^{-1} \mathcal{I}_{12})^{-1}$. Hence, if $\mathcal{I}_{12} = 0$, $\mathcal{I}^{11}$ simplifies to $\mathcal{I}_{11}^{-1}$ - intuitively, $\mathcal{I}_{12}$ says uncertainty about θ_2 estimation adds to uncertainty about θ_1 estimation. If θ_2 known (not estimated), or does not impact θ_1 , then expect no impact on uncertainty of θ_1 estimation. In general, not guaranteed.
 - $\mathcal{I}^{11}(\theta_0)$ can be replaced with either $\mathcal{I}^{11}(\hat{\theta}_n)$ or $\mathcal{I}^{11}(\tilde{\theta}_n)$ under H_n
 - $\mathcal{I}^{11}(\theta_0)$ can be estimated with either $n^{-1}(\nabla \ell)' \nabla \ell$ or $-n^{-1} \nabla^{(2)} \ell$

General Composite Hypothesis

- General vector-valued hypothesis $H_0 : H(\theta_0) = 0$. $\hat{\theta}_n, \tilde{\theta}_n$ similar to before.

$$H_0 : H(\theta_0) = 0 \quad \text{vs.} \quad H_n : H(\theta_0) = \xi/\sqrt{n}$$

$$\hat{\theta}_n = \operatorname{argmax}_{\theta \in \Theta} Q_n(\theta), \quad \tilde{\theta}_n = \operatorname{argmax}_{\theta \in \Theta: H(\theta)=0} Q_n(\theta)$$

- Define Jacobian $\mathcal{J} := \nabla H(\theta_0)$, assume $\operatorname{rank}(\mathcal{J}) = q \leq p$. Denote $\mathcal{I} = \mathcal{I}(\theta_0)$

Theorem 4

AE/HT - III: Under Lehmann conditions,

$$W = nH(\hat{\theta}_n)'(\mathcal{J}'\mathcal{I}^{-1}\mathcal{J})^{-1}H(\hat{\theta}_n)$$

$$LR = 2(\ell(\hat{\theta}_n) - \ell(\tilde{\theta}_n))$$

$$LM = n^{-1}\nabla\ell(\tilde{\theta}_n)'\mathcal{I}^{-1}\mathcal{J}(\mathcal{J}'\mathcal{I}^{-1}\mathcal{J})^{-1}\mathcal{J}'\mathcal{I}^{-1}\nabla\ell(\tilde{\theta}_n)$$

all converge in distribution to $\chi_q^2(\delta)$, where $\delta = \xi'(\mathcal{J}'\mathcal{I}^{-1}\mathcal{J})^{-1}\xi$ (including $H_0 : \xi = 0$).

General Composite Hypothesis – Wald

- By Delta method (i.e. first order Taylor approximation + Slutsky's Theorem)
- Expand $H(\hat{\theta}_n)$ around θ_0

$$\sqrt{n}H(\hat{\theta}_n) = \sqrt{n}H(\theta_0) + \nabla H(\theta_0)\sqrt{n}(\hat{\theta}_n - \theta_0) + o(\sqrt{n}|\hat{\theta}_n - \theta_0|)$$

$\sqrt{n}H(\theta_0) = \xi$ and $o(\sqrt{n}|\hat{\theta}_n - \theta_0|) = o_p(1)$. Since $\nabla H(\theta_0) = \mathcal{J}$, by Slutsky's Theorem,

$$\nabla H(\theta_0)\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{J} \cdot N(0, \mathcal{I}^{-1}) \sim N(0, \mathcal{J}'\mathcal{I}^{-1}\mathcal{J})$$

- Hence $\sqrt{n}H(\hat{\theta}_n) \xrightarrow{d} N(\xi, \mathcal{J}'\mathcal{I}^{-1}\mathcal{J})$
- Use $Z'\Omega^{-1}Z \sim \chi_q^2(\delta)$ with $\delta = \xi'\Omega^{-1}\xi$ when $Z \sim N(\xi, \Omega)$ and replace $\Omega = \mathcal{J}'\mathcal{I}^{-1}\mathcal{J}$

General Composite Hypothesis – LR

- Expand $\ell(\tilde{\theta}_n)$ around $\hat{\theta}_n$, note from before $\nabla \ell(\hat{\theta}_n) = 0$, and $-\frac{1}{n}\nabla^2 \ell(\hat{\theta}_n) \xrightarrow{p} \mathcal{I}$:

$$\begin{aligned} 2(\ell(\hat{\theta}_n) - \ell(\tilde{\theta}_n)) &= -(\hat{\theta}_n - \tilde{\theta}_n)' \nabla \ell(\hat{\theta}_n) - (\hat{\theta}_n - \tilde{\theta}_n)' \nabla^2 \ell(\hat{\theta}_n) (\hat{\theta}_n - \tilde{\theta}_n) \\ &= n(\hat{\theta}_n - \tilde{\theta}_n)' \mathcal{I} (\hat{\theta}_n - \tilde{\theta}_n) + o_p(1) \end{aligned}$$

- Now we would like to relate this to Wald...
- First, Lagrangian for $\tilde{\theta}_n$ (γ : Lagrange multiplier), and FOCS:

$$\mathcal{L}_n(\theta, \gamma) = \ell(\theta) - \gamma' H(\theta), \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{n} \nabla \ell(\tilde{\theta}_n) - \nabla H(\tilde{\theta}_n) \sqrt{n} \tilde{\gamma}_n \\ -H(\tilde{\theta}_n) \end{bmatrix}$$

- It can be shown that $\tilde{\theta}_n \xrightarrow{p} \theta_0$ (won't show this, but show asymptotic distribution)

General Composite Hypothesis – LR under null

- Under the null, $H(\theta_0) = 0$, FOC expansion around θ_0 ,

$$\nabla \ell(\tilde{\theta}_n) = \nabla \ell(\theta_0) + \nabla^2 \ell(\theta_0)(\tilde{\theta}_n - \theta_0); \quad \nabla H(\tilde{\theta}_n) \xrightarrow{p} \mathcal{J}; \quad H(\tilde{\theta}_n) = \mathcal{J}(\tilde{\theta}_n - \theta_0) + o_p(1)$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{n} \nabla \ell(\theta_0) + \nabla^2 \ell(\theta_0) \sqrt{n}(\tilde{\theta}_n - \theta_0) - \mathcal{J} \sqrt{n} \tilde{\gamma}_n \\ -\mathcal{J}(\tilde{\theta}_n - \theta_0) \end{bmatrix} + o_p(1)$$

- Rearranging, and use partition matrix inverse result

$$\begin{bmatrix} \sqrt{n}(\tilde{\theta}_n - \theta_0) \\ \frac{1}{\sqrt{n}} \tilde{\gamma}_n \end{bmatrix} = \begin{bmatrix} \mathcal{I} & \mathcal{J} \\ \mathcal{J} & 0 \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{\sqrt{n}} \nabla \ell(\theta_0) \\ 0 \end{bmatrix} + o_p(1)$$

- $\sqrt{n}(\tilde{\theta}_n - \theta_0) = (\mathcal{I}^{-1} - \mathcal{I}^{-1} \mathcal{J}(\mathcal{J}' \mathcal{I}^{-1} \mathcal{J})^{-1} \mathcal{J}' \mathcal{I}^{-1}) \frac{1}{\sqrt{n}} \nabla \ell(\theta_0)$ then
 $\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_0) = \mathcal{I}^{-1} \mathcal{J}(\mathcal{J}' \mathcal{I}^{-1} \mathcal{J})^{-1} \mathcal{J}' \mathcal{I}^{-1} \frac{1}{\sqrt{n}} \nabla \ell(\theta_0)$

General Composite Hypothesis – LR under null

- From earlier

$$\begin{aligned}
 2(\ell(\hat{\theta}_n) - \ell(\tilde{\theta}_n)) &= n(\hat{\theta}_n - \tilde{\theta}_n)' \mathcal{I}(\hat{\theta}_n - \tilde{\theta}_n) + o_p(1) \\
 &= \frac{1}{\sqrt{n}} \nabla \ell(\theta_0)' \mathcal{I}^{-1} \mathcal{J} (\mathcal{J}' \mathcal{I}^{-1} \mathcal{J})^{-1} \mathcal{J}' \mathcal{I}^{-1} \mathcal{I} \mathcal{I}^{-1} \mathcal{J} (\mathcal{J}' \mathcal{I}^{-1} \mathcal{J})^{-1} \mathcal{J}' \mathcal{I}^{-1} \frac{1}{\sqrt{n}} \nabla \ell(\theta_0) \\
 &= \frac{1}{\sqrt{n}} \nabla \ell(\theta_0)' \mathcal{I}^{-1} \mathcal{J} (\mathcal{J}' \mathcal{I}^{-1} \mathcal{J})^{-1} \mathcal{J}' \mathcal{I}^{-1} \frac{1}{\sqrt{n}} \nabla \ell(\theta_0)
 \end{aligned}$$

- $\frac{1}{\sqrt{n}} \nabla \ell(\theta_0) = \frac{1}{\sqrt{n}} \nabla \ell(\hat{\theta}_n) + \frac{1}{\sqrt{n}} (\theta_0 - \hat{\theta}_n) \nabla^2 \ell(\hat{\theta}_n) = \sqrt{n}(\hat{\theta}_n - \theta_0) \left(-\frac{1}{n} \nabla^2 \ell(\hat{\theta}_n)\right) \xrightarrow{d} N(0, \mathcal{I})$
- Complicated version of $Z' \Omega^{-1} Z$, where $Z \sim N(0, \Omega)$

General Composite Hypothesis – LR under alternative

- Null implied $H(\theta_0) = 0$. For alternative, use $\sqrt{n}H(\theta_0) = \xi$
- This is left as an exercise (please do it!)

General Composite Hypothesis – LM under null

- First, $n^{-1/2}\nabla\ell(\tilde{\theta}_n) = \frac{1}{n}\nabla^2\ell(\tilde{\theta}_n)\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) + o_p(1) \xrightarrow{p} \mathcal{I} \cdot \sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n)$. Then,

$$\begin{aligned} LM &= n^{-1}\nabla\ell(\tilde{\theta}_n)'\mathcal{I}^{-1}\mathcal{J}(\mathcal{J}'\mathcal{I}^{-1}\mathcal{J})^{-1}\mathcal{J}'\mathcal{I}^{-1}\nabla\ell(\tilde{\theta}_n) \\ &= \sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n)'\mathcal{I}\mathcal{I}^{-1}\mathcal{J}(\mathcal{J}'\mathcal{I}^{-1}\mathcal{J})^{-1}\mathcal{J}'\mathcal{I}^{-1}\mathcal{I}\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n) \end{aligned}$$

- Earlier, $\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_0) = AB^{-1}A'\frac{1}{\sqrt{n}}\nabla\ell(\theta_0) + o_p(1)$ where $A = \mathcal{I}^{-1}\mathcal{J}$, $B = \mathcal{J}'\mathcal{I}^{-1}\mathcal{J}$

$$\begin{aligned} LM &= \frac{1}{\sqrt{n}}\nabla\ell(\theta_0)'AB^{-1}A'\mathcal{J}B^{-1}\mathcal{J}'AB^{-1}A'\frac{1}{\sqrt{n}}\nabla\ell(\theta_0) \\ &= \frac{1}{\sqrt{n}}\nabla\ell(\theta_0)'AB^{-1}BB^{-1}BB^{-1}A'\frac{1}{\sqrt{n}}\nabla\ell(\theta_0) = \frac{1}{\sqrt{n}}\nabla\ell(\theta_0)'AB^{-1}A'\frac{1}{\sqrt{n}}\nabla\ell(\theta_0) \end{aligned}$$

- This is equal the expression we had for LR

General Composite Hypothesis – LM under null

- To link to Lagrange multipliers,

$$\tilde{\theta}_n = \underset{\theta}{\operatorname{argmin}} \{ \ell(\theta) - \lambda' H(\theta) \}$$

- FOCs:

$$\nabla \ell(\tilde{\theta}_n) = \nabla H(\tilde{\theta}_n) \lambda = \mathcal{J}(\tilde{\theta}_n) \lambda$$

- Plug this into the LM test, and use $\mathcal{J} = \mathcal{J}(\tilde{\theta}_n)$, $\mathcal{I} = \mathcal{I}(\tilde{\theta}_n)$,

$$LM = n^{-1} \lambda \mathcal{J}' \mathcal{I}^{-1} \mathcal{J} \lambda$$